

Curriculum Vitae

MATTIA PUDDU

Date and place

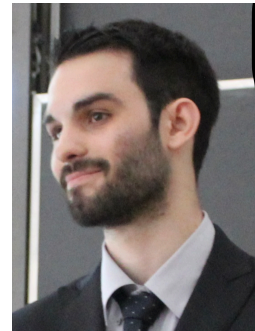
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Employment

2025 – **PhD Student in Mathematics**, RWTH Aachen University, Germany
Project: Effective Difference Algebra and Difference Algebraic Groups
Advisors: Prof. Daniel Robertz and Priv.-Doz. Dr. Annette Bachmayr

Education

2019 – 2025 University of Pisa, **Master’s Degree in Mathematics**.
Final grade: 110/110 *summa cum laude*
Thesis: *The Differential Galois Group of the Family of All Regular Singular Differential Equations over $\mathbb{C}(z)$.*

Advisor: Prof. Tamás Szamuely

Examiner: Prof. Andrea Maffei

Available at: <https://mattiapuddu25.github.io/MasterThesis.html>

2015 – 2019 University of Pisa, **Bachelor’s Degree in Mathematics**.
Final grade: 110/110 *summa cum laude*.
Thesis: *The Picard-Vessiot Theory – An Introduction with Applications to Fuchsian Differential Equations.*

Advisor: Prof. Paolo Acquistapace.

Available at: <https://mattiapuddu25.github.io/BachelorThesis.html>

2010 – 2015 Scientific High School “Enrico Fermi”, Nuoro, Italy.
Final grade: 100/100 *summa cum laude*.

Conferences, Workshops and Summer Schools

May 2026 DART 2026 — Differential Algebra and Related Topics XIV.
CNRS Center La Vieille Perrotine, Île d’Oléron, France, May 24–30, 2026.
Poster: Normal form algorithm for polynomial membership and quasi-simple difference systems.
<https://magix.lix.polytechnique.fr/magix/odelix/dart26/dart26.html>

Sep 2025 Summer School “Computer Algebra with OSCAR”.
Pfalz Akademie, Lambrecht, Germany, September 15–19, 2025.
<https://www.oscar-system.org/events/meetings/2025-09/>

Teaching activity

- 2023 University of Pisa — Department of Civil and Industrial Engineering
Tutor for the pre-course *Matematica 0* (Prof. Irene Venturi)
- Held weekly online sessions to help students pass the OFA admission tests
- 2022 University of Pisa — Department of Agricultural, Food and Agro-Environmental Sciences
Teaching assistant for *Matematica e Statistica* (Bachelor's in Viticulture and Oenology, Prof. Valentino Magnani)
- Conducted the “Mathematics 0” pre-course, including preparation of exams and corresponding solutions
- Held weekly in-class meetings with students for problem-solving and theoretical clarifications
- Reviewed exam papers and solutions from midterm tests throughout the semester
- Supervised students during the exams mentioned above
- 2021 – 2022 University of Pisa — Department of Computer Science
Tutor for *Algebra Lineare B* (Bachelor's in Computer Science, Prof. Mauro Di Nasso)
- Held meetings with students (mostly upon request) for problem-solving and theoretical clarifications
- 2019 – 2020 University of Pisa — Department of Physics
Teaching assistant for *Geometria A* (Bachelor's in Physics, Profs. Mario Salvetti and Filippo Di Santo)
- Held meetings with students (mostly upon request) for problem-solving and theoretical clarifications
- Corrected midterm and final exams, including the preparation of written solutions

Computer Skills

Operating systems: Windows, macOS, Linux, Android

Software/languages: Matlab, C, Julia, HTML (basic), L^AT_EX (expert)

Language Skills

Italian: native

English: fluent

Spanish: independent user

Selected Elective Courses

Completed during the Master's and Bachelor's degrees at the University of Pisa, in addition to the mandatory curriculum.

1. **Coxeter Groups** (geometric realization and combinatorial characterization; Bruhat order; root systems; classification of finite Coxeter groups; applications in combinatorics and topology) with a grade of 30/30 *cum laude*
(taught by Profs. Michele D'Adderio and Mario Salvetti)
2. **Elements of Algebraic Geometry** (projective geometry; Bezout's theorem; Nullstellensatz; affine and quasi-projective varieties; Zariski topology; irreducible decomposition; morphisms and rational maps; dimension; tangent spaces and singular points; Segre, Veronese, and Grassmannian varieties) with a grade of 30/30 *cum laude*

(taught by Prof. Rita Pardini)

3. **Algebraic Geometry C** (Riemann surfaces; holomorphic and meromorphic functions; group actions and Hurwitz bound; differential forms; divisors and linear systems; Riemann-Roch theorem; Serre duality; Weierstrass points; Jacobian variety; theorem of Abel and Jacobi) with a grade of 30/30 *cum laude*
(taught by Prof. Marco Franciosi)
4. **Algebraic Combinatorics** (enumerative and algebraic combinatorics; ordinary and exponential generating functions; Lagrange inversion; symbolic methods; transfer matrix method; MacMahon master theorem; symmetric functions) with a grade of 30/30 *cum laude*
(taught by Prof. Michele D'Adderio)
5. **Analytic Number Theory A** (entire functions of finite order; Gamma function and Stirling's formula; Riemann Zeta function; Riemann-von Mangoldt formula; prime number theorem and de la Vallée Poussin theorem; Mellin transform; characters and Dirichlet L -functions; primes in arithmetic progressions) with a grade of 30/30 *cum laude*
(taught by Prof. Giuseppe Puglisi)
6. **Groups and Representations** (representations of finite groups; complete reducibility; Schur's lemma; characters and orthonormality; induced representations; representations of semisimple algebras; representations of the symmetric group; Schur's duality; representations of the general linear group) with a grade of 30/30 *cum laude*
(taught by Profs. Michele D'Adderio and Giovanni Gaiffi)
7. **Elements of Algebraic Topology** (singular homology; CW-complexes; cellular homology; cohomology ring and cup product; Poincaré duality; homotopy groups; cellular approximation; homotopy groups of spheres) with a grade of 30/30 *cum laude*
(taught by Prof. Filippo Callegaro)
8. **Elements of Complex Analysis** (Hurwitz theorems; Schwarz lemma; automorphisms of the disk and the Riemann sphere; Poincaré distance; Wolff-Denjoy theorem; Riemann uniformization theorem; several complex variables; Hartogs extension theorem; plurisubharmonic functions; Levi problem) with a grade of 30/30 *cum laude*
(taught by Prof. Marco Abate)
9. **Modular Forms** (the modular group $SL_2(\mathbb{Z})$; classical modular forms; Fourier expansions; Eisenstein series; theta series and representations as sums of squares; congruence subgroups; Hecke operators; Petersson inner product; L -functions attached to eigenforms; Euler product decomposition) with a grade of 30/30 *cum laude*
(taught by Profs. Andrea Maffei and Davide Lombardo)
10. **Lie Algebras and Lie Groups** (semisimple, nilpotent, and solvable Lie algebras; Killing form; Weyl's theorem; representations of $\mathfrak{sl}(2, \mathbb{K})$; weight lattice; root systems; Dynkin diagrams and classification; Poincaré-Birkhoff-Witt theorem; Serre relations; Lie groups; Baker-Campbell-Hausdorff formula) with a grade of 30/30 *cum laude*
(taught by Profs. Giovanni Gaiffi and Enrico Le Donne)
11. **Algebraic Number Theory 1** (integral extensions; trace and norm; ring of integers; Dedekind domains; fractional ideals and class group; Kummer's theorem; ramification and inertia in Galois extensions; decomposition and inertia groups; Frobenius morphism; Minkowski convex body theorem; finiteness of the class group; Dirichlet unit theorem) with a grade of 28/30
(taught by Profs. Ilaria Del Corso and Roberto Dvornicich)
12. **Elementary Number Theory** (prime number theorem; Lagrange's four-squares theorem; quadratic residues and reciprocity law; multiplicative groups modulo q ; arithmetic functions; Dirichlet series; Mertens and Chebychev theorems; Dirichlet's theorem on primes in arithmetic progressions) with a grade of 30/30 *cum laude*
(taught by Prof. Giuseppe Puglisi)

Other Activities

Mathematical Olympiads. I have always been interested in the Italian Mathematical Olympiad project. Over the years, I compiled my solutions to the National Team Competitions held in Cesenatico into two files, which also include theory notes useful for competition preparation.

Available at: <https://mattiapuddu25.github.io/MathOlympiads.html>